Do CODI-GPT-2 and CODI-Llama actually use latent thoughts?

Three tests, run on both models

- Force-decode per step: how many right↔wrong flips per latent step?
- LM-head probe: how stable is the top-1 token across steps?
- Distribution probe: cos sim / KL between softmax distributions of consecutive steps

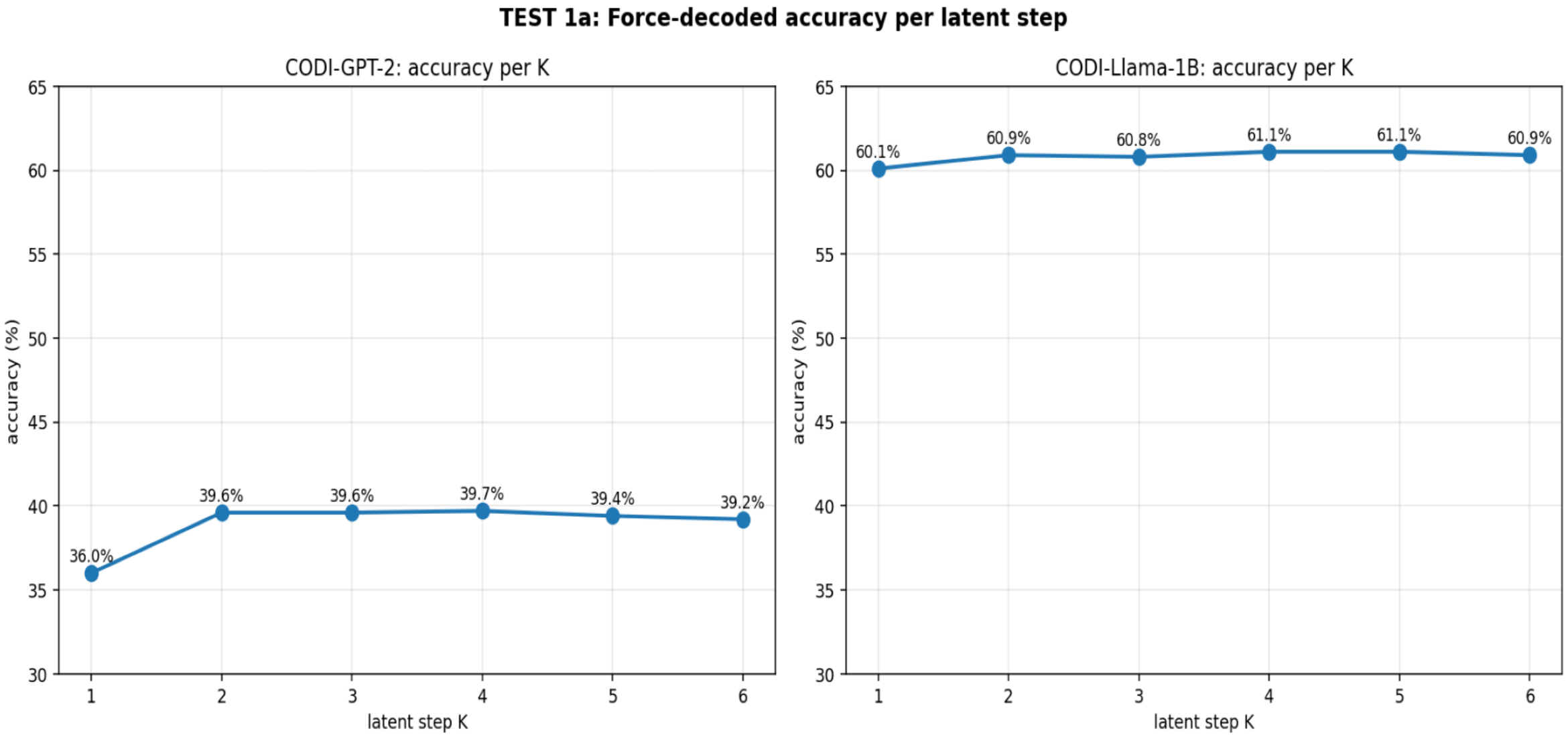
What the user expected

Hypothesis: CODI-Llama doesn't really use latents — it decides early.

Setup

1000 SVAMP problems (GPT-2) / 1000 (Llama); K=6 (trained value).

TEST 1a · Force-decoded accuracy per latent step

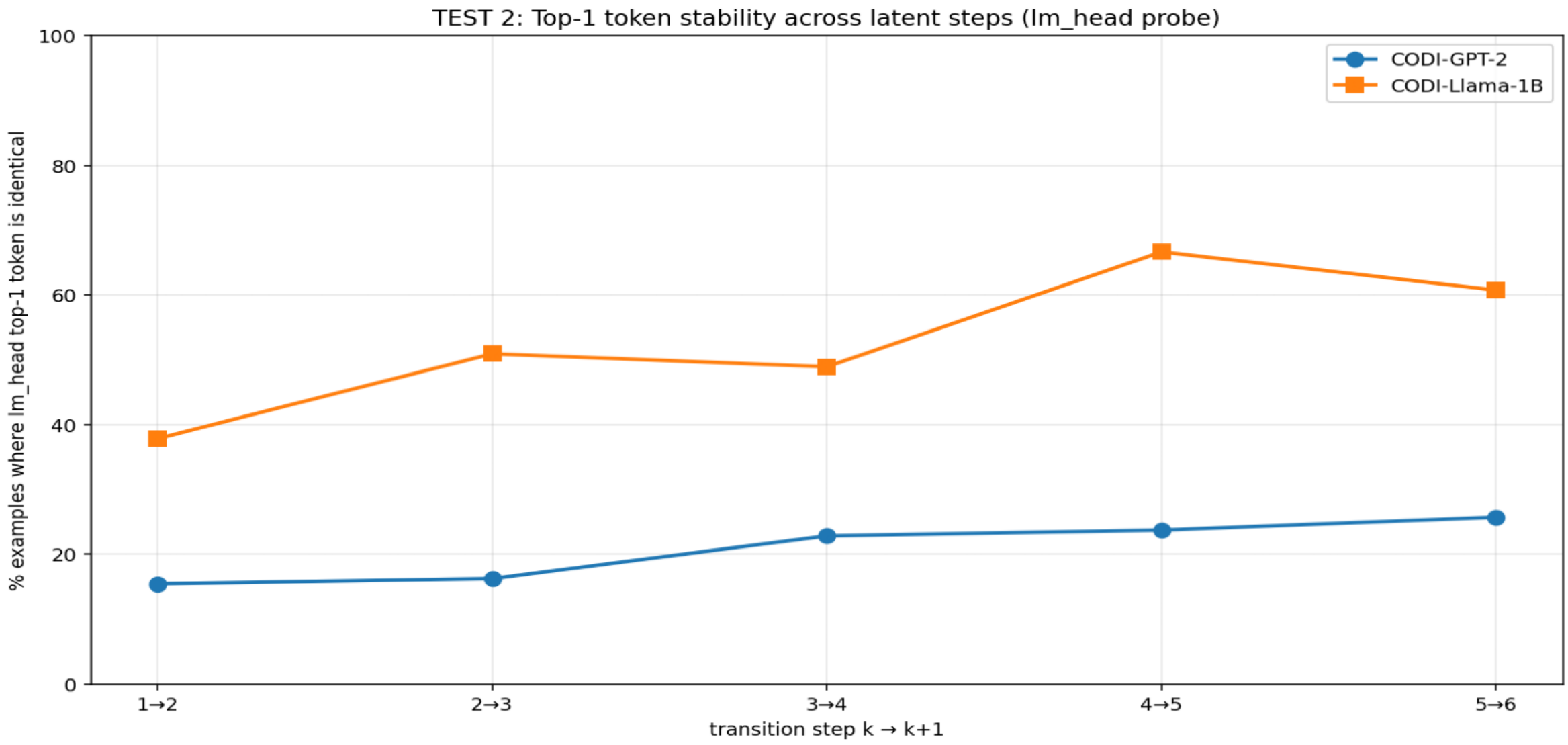


Llama: step 1 = 60.1%, peak = 61.1% (range 1.0 pp). GPT-2: step 1 = 36.0%, peak = 39.7% (gain +3.7 pp). Llama's latent loop adds little to accuracy; GPT-2 climbs early then plateaus.

TEST 1b · Per-step transitions (right→wrong vs wrong→right)

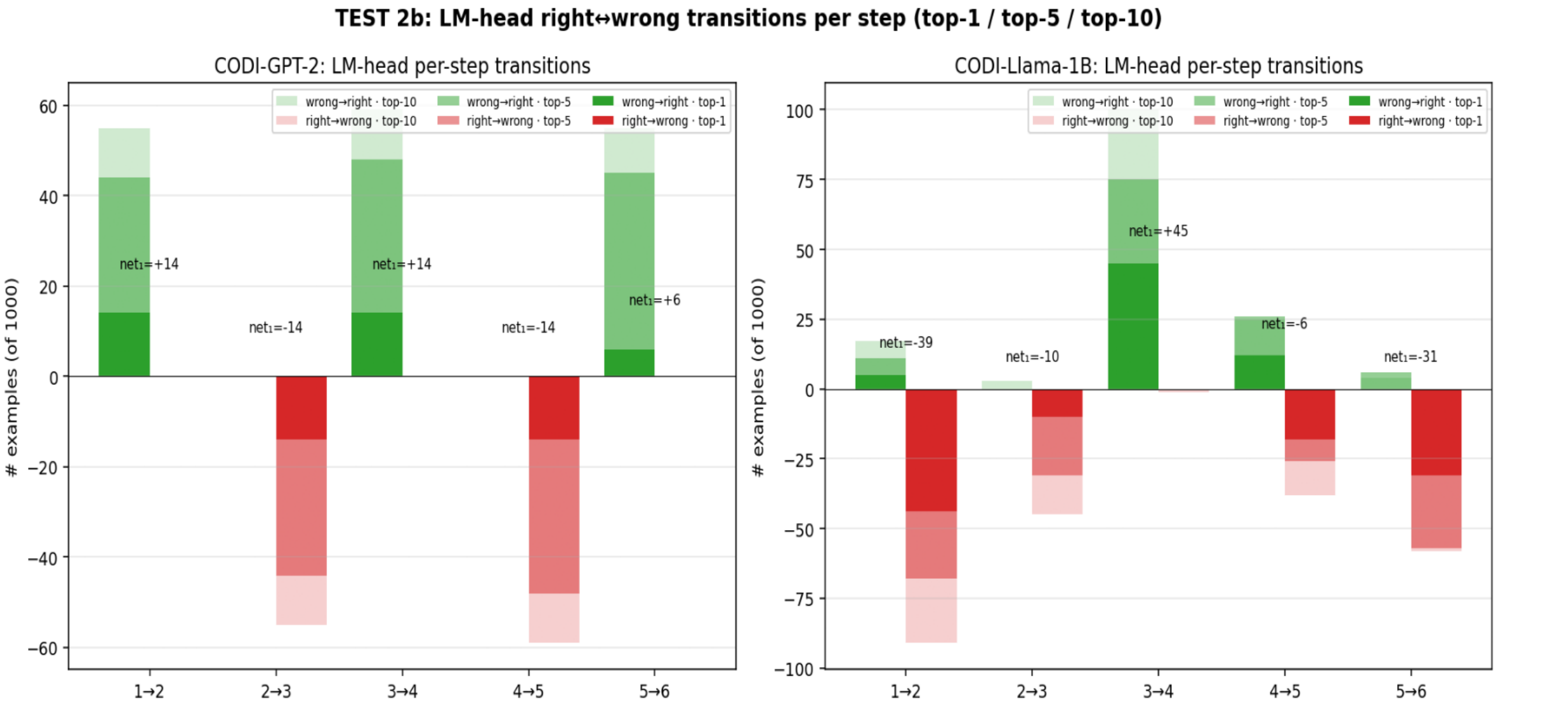
GPT-2 net flips per step: [36, 0, 1, -3, -2] (out of 1000). Llama: [8, -1, 3, 0, -2] (out of 1000). Where net ≈ 0 the latent loop is doing no useful work on net.

TEST 2 · LM-head top-1 token stability per step transition



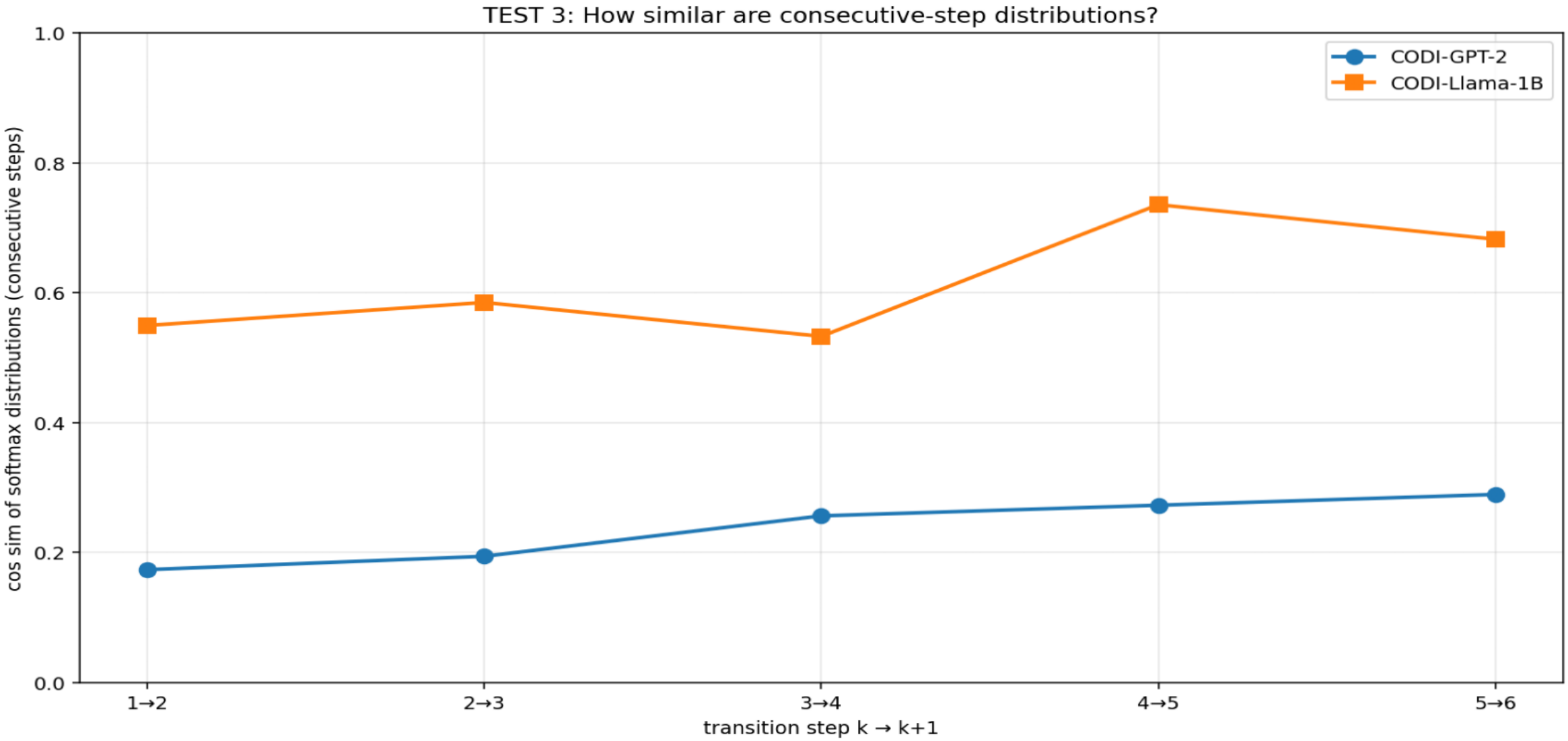
GPT-2 top-1 stability: 15-26%. Llama: 38-67%. Higher stability = less work being done across the step transition.

TEST 2b · LM-head transitions (right→wrong / wrong→right)



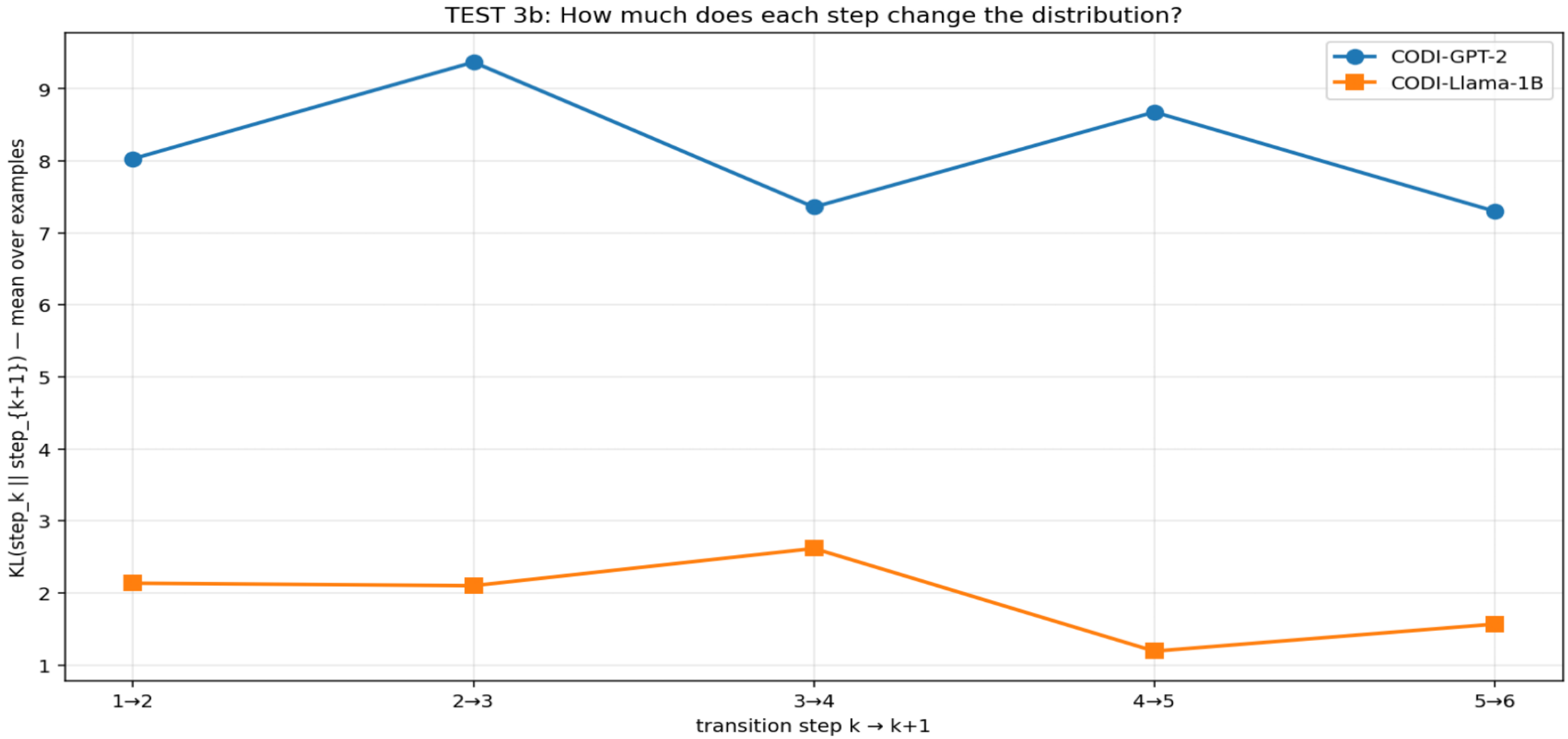
p k = LM-head top-1 token equals the first BPE token of " {gold}". Solid = top-1, translucent = top-5, faintest = top-10. GPT-2 top-1 acc 0.0-1.4% (nets [14, -14, 14, -14, 6]); Llama top-1 acc 0.0-4.9% (nets [-39, -1

TEST 3 · Distribution similarity (cos sim) across latent steps



GPT-2 cos sim range: 0.17-0.29. Llama: 0.53-0.74. Higher cos sim = consecutive distributions are more alike (less change per step).

TEST 3b · KL divergence between consecutive distributions



GPT-2 KL range: 7.30-9.37 nats. Llama: 1.19-2.62 nats. Smaller KL = less distributional change per step.

Synthesis: do they use the latents?

Setup: 1000 SVAMP problems (GPT-2) / 1000 (Llama), K=6.

CODI-GPT-2

- Step 1→2 net flips: +36 (wrong→right=70, right→wrong=34) out of 1000.
- Other transitions in K=2..6: |net| ≤ 3 (drift, not progress).
- Accuracy step 1 → peak: 36.0% → 39.7% (gain +3.7 pp).
- Top-1 LM-head token stability: 15-26% — high volatility.
- Cos sim 0.17-0.29, KL 7.3-9.4 nats — distributions rotate a lot,

but the rotation mostly cancels out behaviorally.

CODI-Llama-1B

- Accuracy step 1 → step 6: 60.1% → 60.9% (range 1.0 pp).
- All transitions in K=1..6: |net| ≤ 8 out of 1000.
- Top-1 LM-head token stability: 38-67% — high stability.
- Cos sim 0.53-0.74, KL 1.2-2.6 nats — latents barely change.
- Effectively at 'final' accuracy from step 1 of the latent loop.

Conclusion

Hypothesis CONFIRMED. CODI-Llama doesn't meaningfully use its latent thoughts — the answer is determined by the prompt + first latent step. Subsequent steps are essentially decorative.

CODI-GPT-2 does have ONE step of genuine computation (step 1→2), but later steps don't accumulate useful work either.